

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Potato Zero Tillage and Mulching (PZTM) in India

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.1 · Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Version 1.0 · June 2026

Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

In the eastern Indo-Gangetic Plains, particularly Bihar, intensive rice-wheat systems have led to degrading soils, falling water tables, and the pervasive burning of crop residues, which worsens air quality and contributes to greenhouse gas emissions. Smallholder farmers, the primary beneficiaries of this initiative, face erratic rainfall, rising input costs, and labor shortages that lock them into low productivity and high-risk farming. Women, who provide the majority of farm labor, are disproportionately affected, shouldering the heavy drudgery of planting and harvesting while having limited access to technology and decision-making power.

This solution track aims to transform conventional potato cultivation by scaling **Potato Zero Tillage and Mulching (PZTM)** across the Indo-Gangetic Plains, starting with Bihar and expanding to coastal areas. PZTM is a validated innovation bundle where early-maturing potato seed tubers are placed on unplowed rice fallow soil and covered with a thick layer of rice straw mulch. This practice enables immediate planting, conserves soil moisture, suppresses weeds, reduces production costs by 25–30%, and eliminates the need for residue burning. The bundle is delivered through a proven, gender-responsive model that combines Digital Green's video-mediated extension with the vast network of JEEViKA women's self-help groups (SHGs).

With over five years of research and piloting, including a randomized controlled trial (RCT) involving 2,000 households in Bihar, the evidence base for PZTM is robust. Results show a 16% increase in gross margins, a 30–40% reduction in pest and disease incidence, and significant gains in women's decision-making power. This S4I investment will act as catalytic capital to transition from a successful pilot to a systemic scaling effort, leveraging public extension systems, fostering public-private partnerships, and institutionalizing PZTM within state government programs.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **Potato Zero Tillage and Mulching (PZTM):** A conservation agriculture practice where early-maturing potato varieties are planted directly into rice residue and mulched with rice straw.
- **Digital and Community Extension Model:** Video-mediated learning content delivered through women's SHGs to drive behavior change and adoption.
- **Institutional and Market Linkages:** Integration into state extension manuals, development of mulch supply systems, and private sector engagement. Certification mechanisms to link farmers to carbon markets and private sector buyers will be explored.

2.2 Evidence of Impact

- **Profitability:** RCT data from Bihar shows a **16% higher gross margin** for farmers, driven by a **31% reduction in production costs** (labor, fuel, irrigation) and a **5.4% increase in marketable yields**.
- **Agronomic Gains:** On-farm trials demonstrate a **30–40% reduction in pest and disease incidence** compared to conventional tillage.
- **Water Efficiency:** PZTM reduces irrigation events by up to **50%**, doubling water productivity.
- **Climate and Environmental:** Eliminates residue burning, reducing GHG emissions by an estimated **0.72 tons CO₂e per hectare**.
- **Gender and Inclusion:** The intervention has led to increased joint decision-making (65% of adoption decisions are made jointly by men and women) and reduced drudgery for women in weeding and harvesting.

2.3 Scaling Targets

Outcome Metric		2026 Target	Long-term Goal (3–5 years)
Farmers reached (cumulative)		10,000 households	50,000+ households (50% women)
Area under PZTM (ha)		5,000 ha	10,000+ ha
Women in leadership roles		250 promoters	1,000+ women promoters
Investment leveraged		\$500,000	\$2,000,000+ (state and private sector)

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The scaling pathway is designed to transition PZTM from a project-led intervention to a publicly owned and privately supported agricultural practice. The strategy operates through three interconnected channels:

- **Institutional Anchoring (Public Sector):** The primary pathway is integrating PZTM into the Government of Bihar's extension system. By embedding the practice into state agricultural roadmaps, training curricula, and extension officers' mandates, we ensure long-term sustainability and reach. The Department of Agriculture and Horticulture will chair a State PZTM Coordination Committee to oversee this mainstreaming.
- **Grassroots Delivery (NGO and Community):** We will continue leveraging the JEEViKA network of women's SHGs, facilitated by Digital Green's video-mediated learning, to drive adoption at the last mile. This channel ensures that information is trusted, localized, and reaches marginalized farmers, particularly women.
- **Market-Based Scaling (Private Sector and PPP):** We are cultivating partnerships with private sector actors (PepsiCo, McCain, SAI Platform) to integrate PZTM into their supply chains. By demonstrating the business case (lower carbon footprint, more uniform tubers), we aim to create pull mechanisms that incentivize adoption without ongoing CGIAR facilitation.

3.2 Theory of Change

If smallholder farmers in Bihar gain access to the PZTM innovation bundle through trusted community networks (JEEViKA/Digital Green) and government extension services, **and** key systemic barriers—such as straw availability, mechanization gaps, and lack of institutional recognition—are addressed through multi-stakeholder partnerships and policy engagement, **then** farmers will adopt the practice at scale, leading to reduced production costs, higher and more stable incomes, and improved soil health. This will **contribute to** a systemic shift toward climate-smart agriculture in the Indo-Gangetic Plains, with women playing a central role, and position PZTM as a model for sustainable intensification across South Asia.

3.3 Key Activities Funded by S4I

- **Strategic Prioritization and Co-design:** Convene “harvesting day” with Bihar Minister of Agriculture, a multi-stakeholder workshop with government, private sector, and NGOs to finalize a focused scaling strategy and prioritize the most impactful activities for 2026.

- **Institutional Anchoring:** Contract a consultant (ideally seconded from the Ministry) to support the integration of PZTM into state extension manuals and facilitate the submission of a state-funded scaling proposal. Training of extension officers is also foreseen.
- **Delivery Innovation:** Deployment of a nature-inspired and targeted approach. 6-8 PZTM videos will be updated, youth involved in mulch coordination, and Farmer.Chat AI tool will be used to provide real-time agronomic advice.
- **Market and Finance Exploration:** Conduct a feasibility study for a PZTM certification system and registry (using AI and NFTs) to explore links to carbon credit markets and government subsidy schemes.

4. Partners and Roles

Partner Type	Specific Partners and Quantities		Role in Scaling Pathway
Institutional and Scaling Partners	Government of Bihar (Dept. of Agriculture and Horticulture); Bihar Rural Livelihoods Promotion Society (JEEViKA)		Government: Chairs the state coordination committee, mainstreams PZTM into extension programs, and provides institutional backing and funding pathways. JEEViKA: Provides last-mile access to women farmers through its network of SHGs and Village Resource Persons.
Knowledge and Delivery Partners	Digital International Center (CIP)	Green; Potato	Digital Green: Leads video-mediated extension, develops AI tools (), and trains community facilitators. CIP: Provides technical leadership, agronomic backstopping, and MELIA.
Demand and Market Partners	PepsiCo, McCain, SAI Platform; Farmer Producer Organizations (FPOs)		Private Sector: Explore integration of PZTM into contract farming schemes, provide market pull for certified produce, and co-invest in scaling activities. FPOs: Aggregate produce and link farmers to markets.

4.4 Funding Model: S4I catalytic funding covers strategic coordination, partnership development, and innovation testing (e.g., certification pilots, AI tools), while the core cost of adoption is borne by farmers through reduced input expenses. State government budgets and private sector investment will be leveraged to finance large-scale training, demonstrations, and mechanization subsidies post-2026.

5. Enabling Environment and Risks

5.1 Opportunities

- **Strong Government Alignment:** PZTM directly supports the Bihar State Agricultural Roadmap (2023-2028), which prioritizes climate-smart agriculture and residue management. The Department of Agriculture has formally invited a scaling proposal, signaling readiness to invest state funds.
- **Private Sector Interest:** Processors like PepsiCo and McCain are actively seeking to decarbonize their supply chains and have expressed interest in sourcing low-carbon potatoes from PZTM farmers.

- **Digital Infrastructure:** The existence of Digital Green's platform and the spread of smartphones in rural areas create an opportunity to test AI-enabled extension (Farmer.Chat) and digital certification systems at low cost.
- **Regional Replicability:** Agronomic trials confirm PZTM works in maize-based systems in Africa, opening a long-term pathway for cross-continental scaling.

5.2 Challenges and Mitigation

- **Straw Availability and Trade-offs:** Rice straw is a critical livestock feed. **Mitigation:** Pilot "straw banks" and paid straw services (youth micro-enterprises); test alternative mulches (maize stover); and promote synchronized community-level mulching to manage resources collectively.
- **Lack of Focus and Prioritization:** The current plan contains too many activities, risking dilution of effort. **Mitigation:** Conduct a stakeholder prioritization workshop in Q2 2026 to funnel the broad list of ideas into a focused set of transformative actions, using evidence from Digital Green's experience to guide decisions.
- **Mechanization Gap:** The current manual model is labor-saving but may not appeal to larger farmers. **Mitigation:** Parallel investment in prototyping a zero-till potato planter (attachable to rice combines) to prepare for mechanized scaling in later phases, while focusing S4I resources on the manual model for smallholders.
- **Gender Risk of Male Takeover:** If PZTM becomes highly profitable, men may take control of the enterprise. **Mitigation:** Maintain explicit targeting of women for training and promoter roles; feature women champions in all communications; and track decision-making dynamics in MELIA.

6. Learning and Adaptation

6.1 Monitoring Focus

- **Adoption and Reach:** Who is adopting (disaggregated by gender, caste, landholding), on what scale (area), and at what rate? (Data collected via the AI-enabled voice survey system as part of rigorous MEL)
- **Behavior Change and Outcomes:** Changes in yield, production costs, and labor allocation; shifts in decision-making power within households.
- **Delivery Effectiveness:** Comparative cost-effectiveness of video-mediated extension vs. AI-enabled extension (Farmer.Chat); partner coordination effectiveness.
- **Systemic Change:** Progress on institutional integration (e.g., PZTM in extension manuals); evidence of private sector co-investment; policy shifts.

6.2 Adaptive Management

- **Quarterly Check-ins:** Regular strategy reviews with the S4I AOW2 team to assess progress against the focused action plan and adjust activities as needed.
- **Mid-Year Review (Q3 2026):** A formal review of adoption data and delivery effectiveness will inform a course correction before the main planting season.
- **GenderUp Check-ins:** Pre- and post-season facilitated sessions to identify and address any unintended negative social consequences of scaling.

7. Next Steps and Timeline (2026)

Quarter	Key Activities
Q1 (Jan-Mar)	Scoping and Prioritization: Contract key partners; convene multi-stakeholder workshop (Govt, Digital Green, JEEViKA, private sector) to finalize focused scaling strategy; conduct stakeholder/political economy analysis to refine partnerships.
Q2 (Apr-Jun)	Content and Capacity Development: Delivery Innovation (targeted outreach, nature-inspired scaling strategy) deployed. Including update 6-8 PZTM video modules; fine-tune AI content; initiate discussions with Bihar government on proposal and manual integration; begin training of master trainers (extension officers, JEEViKA staff).
Q3 (Jul-Sep)	Pre-planting Mobilization: Roll out community video screenings through SHGs; establish side-by-side demonstration plots in target blocks; initiate feasibility study for PZTM certification system; mid-year review to assess progress and adjust.
Q4 (Oct-Dec)	Adoption and Planting: Farmers make planting decisions; implement rapid adoption study to track reach and farmer characteristics; support government submission of state funding proposal; document early lessons for 2027 scale-up.

Communications and Knowledge: Key products will include a co-designed Scaling Strategy Report, farmer testimonial videos, policy briefs on institutional integration, and a blog series on the "learning journey" of prioritizing activities. Communications will target government stakeholders, donors, and the private sector to attract co-investment.

8. Contact

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